cascadeur MMD toolkit

1.11 使用教程
User Guide

VMD 骨骼动画导入、FK / IK 导出与 Scene Clean。

VMD skeletal import, FK / IK export and Scene Clean.

适用于 Windows x64 的 Cascadeur 2026.1.3 与 2026.2 系列。先使用工程副本测试短片段，再处理完整动作。

For Cascadeur 2026.1.3 and the 2026.2 series on Windows x64. Test a short clip in a project copy before processing a full motion.



VMD Import / Export / Scene Clean

版本 / Version 1.11 2026-10-23

阅读导航 / Reading guide

先完成安装、兼容性与准备章节；随后按需查阅对应功能。所有文件位置均以包含 cascadeur.exe 的软件安装根目录为起点。

Start with installation, compatibility and preparation, then open the chapter for the feature you need. Installed paths below are relative to the software root containing cascadeur.exe.

01 安装工具集 / Install the toolkit

02 兼容版本与文件位置 / Compatibility and installed files

03 升级与恢复安装 / Update and restore an installation

04 准备角色工程 / Prepare a character project

05 选择目标模型 / Select target models

06 设置时间轴范围 / Set the timeline range

07 导入 VMD 动作 / Import a VMD motion

08 使用 MMD 道具 / Work with MMD props

09 导出骨骼动画 / Export skeletal animation

10 选择 FK 或 IK 导出 / Choose FK or IK export

11 清理场景中的生成 Rig / Clean generated rigs from a scene

12 查找报错原因 / Diagnose an error

13 完成一次完整工作流 / Complete a full workflow

01 安装工具集

Install the toolkit

工具集 1.11 面向 Windows x64 的 Cascadeur 2026.1.3 与 2026.2 系列。完整解压发布包，保持 Install.cmd、Installer.Core.psm1、清单和 payload 相邻。安装器按实际官方脚本接口选择路线，并检查 CPython 3.11、Qt ABI、NumPy 和所需原生导出。

Toolkit 1.11 supports Cascadeur 2026.1.3 and the 2026.2 series on Windows x64. Extract the full package and keep Install.cmd, Installer.Core.psm1, the manifest and payload together. The installer chooses the route from the official scripting interfaces and checks CPython 3.11, Qt ABI, NumPy and required native exports.

1. 保存正在编辑的工作，关闭要安装插件的那一份 Cascadeur。打开解压后的文件夹，双击 Install.cmd。控制台会显示中英文安装说明，并等待输入。

Save your work and close the Cascadeur installation you intend to update. Open the extracted folder and double-click Install.cmd. A console displays bilingual instructions and waits for input.

2. 粘贴 Cascadeur 的安装根目录，然后按 Enter。例如 C:\Program Files\Cascadeur。该文件夹内应直接存在 cascadeur.exe；不要输入 .exe 文件路径、.casc 工程目录或 resources 子目录。粘贴时带有成对双引号也可以。留空按 Enter 表示取消。

Paste the Cascadeur installation root and press Enter, for example C:\Program Files\Cascadeur. This folder must directly contain cascadeur.exe. Do not enter the executable filename, a .casc project folder or the resources subfolder. Surrounding double quotes are accepted. Leave the input blank and press Enter to cancel.

3. 核对显示的目录与待写入数量。如 Windows 请求管理员权限，确认这是本次安装后允许。等待“安装成功 / Installation completed”，记下显示的备份目录。成功、失败或取消后窗口都会保留，按提示按键关闭。

Check the displayed directory and number of files to write. If Windows requests administrator permission, allow this installation after verifying it. Wait for “Installation completed” and keep the displayed backup directory. The console remains open after success, failure or cancellation until you press the requested key.

4. 重新启动正确目录中的 Cascadeur，打开已保存的角色工程。File → Import 下应有 Import VMD，File → Export 下应有 Export VMD。Scene Clean 位于 Scripts 或 Commands → cascadeur MMD toolkit。

Restart Cascadeur from the directory you updated and open a saved character project. Find Import VMD under File > Import and Export VMD under File > Export. Scene Clean is under Scripts or Commands > cascadeur MMD toolkit.

如果电脑有多个 Cascadeur 安装，请确认启动快捷方式指向刚才输入的目录。此发布包不安装 Cascadeur 本体，也不包含模型、动作或额外的 PMX 导入工具。

If you have multiple Cascadeur installations, check that your shortcut launches the directory you entered. This package does not install Cascadeur itself and does not include models, motions or an additional PMX importer.

已测试版本包括 2026.1.3 与 2026.2 系列；这不是运行或安装白名单。其他版本按实际 API、所需符号和 ABI 检查；缺少必要能力时提示具体原因。

Tested versions include 2026.1.3 and the 2026.2 series; these are not an installation or runtime allowlist. Other hosts are checked for actual APIs, required symbols and ABI, with specific errors for missing requirements.

02 兼容版本与文件位置

Compatibility and installed files

2026.2 路线使用宿主已有的完整 PySide6 与官方脚本服务器，并保留本插件的 D3D11 引导。2026.1.3 路线由发布包补齐匹配的 QtCore、QtNetwork、QtQml、Shiboken 和 python3.dll，通过宿主 QML 直接调用插件，不安装 D3D11 代理。两条路线共用同一套导入、导出与窗口逻辑。

The 2026.2 route uses the host full PySide6, official script server and the toolkit D3D11 entry. The 2026.1.3 route adds matching QtCore, QtNetwork, QtQml, Shiboken and python3.dll from the package, and invokes the toolkit directly through host QML. It does not install a D3D11 proxy. Both routes share the same import, export and dialog code.

运行插件无需另装系统 Python、Blender、MMD、Codex 桥接或网络服务。现代宿主使用的 127.0.0.1:8765 是 Cascadeur 自带的本机脚本服务器。安装包不包含模型、动作或 PMX 导入工具。

No separate system Python, Blender, MMD, Codex bridge or network service is needed to run the toolkit. On the modern host, 127.0.0.1:8765 is the local script server supplied by Cascadeur. Models, motions and a PMX model importer are not included.

正式 Python 包 / Product package	resources/scripts/python/scripts/vmd_bridge/
事件钩子 / Scene hook	resources/scripts/python/events/scene_activated/vmd_menu_inject.py
官方命令 / Official commands	resources/scripts/python/commands/mmd_toolkit/
2026.2 引导 / Modern entry	D3D11.dll
2026.1.3 绑定 / Legacy bindings	python/Lib/site-packages/PySide6/ and shiboken6/
2026.1.3 ABI 引导 / Legacy ABI	python3.dll
旧版引用辅助 / Legacy reference helper	resources/scripts/python/scripts/vmd_bridge/native/qt_list_ref.dll

安装成功不能使旧宿主打开新版场景格式。若 2026.1.3 提示 unknown asset type name: materials_link，请使用由兼容版本保存的场景副本；不要反复安装插件尝试修复场景格式。

Installation does not make an old host understand newer scene formats. If 2026.1.3 reports unknown asset type name: materials_link, use a scene copy saved by a compatible host version. Reinstalling the toolkit does not convert the scene format.

安装器不会覆盖旧版中未知、缺失不全或不匹配的 Qt 绑定及 python3.dll。遇到冲突时先保存错误信息，使用原备份恢复旧安装，不要混装两份绑定或复制宿主 Qt DLL。

The installer refuses unknown, incomplete or mismatched bindings and python3.dll on the legacy route. Keep the error and restore the previous installation from its backup. Do not mix binding distributions or copy host Qt DLLs.

03 升级与恢复安装

Update and restore an installation

升级仍使用 Install.cmd。安装器先保全要替换的文件，再部署新文件；内容相同的文件不会重复写入。未知的其他插件不会被批量清除。如果已有无法识别的 D3D11.dll，安装器会停止并指出冲突，不要手动删除它来强行安装。

Use Install.cmd for updates too. The installer backs up files before replacement and leaves identical files untouched. It does not sweep away unknown plugins. If it finds an unrecognized D3D11.dll, it stops with a conflict message. Do not delete that file simply to force the installation.

安装失败时先阅读完整提示。若已经写入文件，安装器会尝试恢复原内容。若出现 rollback incomplete，说明恢复未完成：保留备份和提示，解决文件占用或权限问题后执行恢复；不要把这类结果当作安装成功。

Read the full message if installation fails. If writes have begun, the installer attempts to restore the previous content. “Rollback incomplete” means recovery did not finish. Keep the backup and message, resolve file locks or permissions, then run Restore. This result is not a successful installation.

1. 关闭相应安装目录中的 Cascadeur，双击 Restore.cmd。

Close Cascadeur from the relevant installation and double-click Restore.cmd.

2. 粘贴原安装窗口显示的完整备份目录并按 Enter。默认备份保存在 %LOCALAPPDATA%\cascadeur MMD toolkit\backups 下，每次安装有独立子目录。可将此路径粘贴到资源管理器地址栏查找。

Paste the full backup directory shown by the original installation and press Enter. Backups default to %LOCALAPPDATA%\cascadeur MMD toolkit\backups, with a separate subfolder for each installation. Paste this path into File Explorer’ s address bar to locate it.

3. 等待恢复成功后再启动 Cascadeur。恢复会还原该次安装前的文件，并移除该次新增的受管文件；它不会恢复动画工程。如果之后又独立修改过插件文件，恢复器可能拒绝覆盖，请保留具体错误信息。

Wait for successful recovery before starting Cascadeur. Restore reinstates the files from before that installation and removes managed files introduced by it. It does not restore animation projects. If plugin files were independently modified afterward, recovery may refuse to overwrite them; keep the exact error message.

安装备份与工程备份用途不同。处理动作前仍应复制 .casc 工程；不要删除唯一的工程副本来换回旧插件。

Installation backups and project backups serve different purposes. Copy the .casc project before working on animation; restoring an older plugin cannot recover a deleted project.

04 准备角色工程

Prepare a character project

工具集把 VMD 骨骼动画写入已有的 MMD 角色工程。请先通过你已有的模型转换与绑定流程，把模型、骨架和 Cascadeur 控制器准备好，并保存为 .casc。Import VMD 不是 PMX 模型导入命令，也不会自动为任意网格制作角色 Rig。

The toolkit writes VMD skeletal motion into an existing MMD character project. Use your established model conversion and rigging workflow to prepare the mesh, skeleton and Cascadeur controls, then save a .casc file. Import VMD does not import a PMX model or automatically build a character rig for an arbitrary mesh.

先打开工程副本，检查网格、骨架和控制器属于同一角色。人物需要可用的人体控制绑定；只看见名为 Rig(0) 的节点，不能证明它已经正确绑定。已有存盘路径的工程在追加模型后，可使用当前场景识别结果；全新未保存工程应先保存。

Open a copy of your project and check that the mesh, skeleton and controls belong to the same character. A character needs valid humanoid control bindings; a node named Rig(0) alone does not prove a correct setup. A project with an existing saved path can use live detection after models are appended. Save a completely new project first.

插件优先检查模型真实绑定和骨架结构，并兼容部分日文骨名的 .L / .R 写法。不要为了让名称看起来标准而盲目重命名足 D、捩骨或辅助骨。名称近似并不能代替正确的父子关系、初始姿势和绑定。

The toolkit examines actual bindings and skeleton structure and supports some Japanese bone names written with .L/.R suffixes. Do not blindly rename D bones, twist bones or helpers merely to make names look standard. Similar names cannot replace correct parent relationships, rest poses and bindings.

如果保有与目标角色严格对应的原始 PMX，可将它放在工程目录或其子目录，或 %LOCALAPPDATA%\cascadeur MMD toolkit\models 中供自动匹配。没有 PMX 时会尝试使用场景信息；不要拿其他角色的 PMX 冒充。不同模型的比例、局部轴和附加变换可能使同一 VMD 的视觉效果不同。

If you have the exact original PMX for the target character, place it in the project folder or a subfolder, or in %LOCALAPPDATA%\cascadeur MMD toolkit\models for automatic matching. Without a PMX, the toolkit attempts to use scene information. Do not substitute another character’ s PMX. Differences in proportions, local axes and inherited transforms can change the appearance of the same VMD.

人物精度导入可能补全或重建该角色的生成控制图。首次迁移旧 Rig 要求场景中只有该角色，并从目标第0帧替换全部已有动画；已有完整独立图才可使用相应的部分区间或多模型作用域。叠加动画、原生循环和混入无关对象的共享层不支持。请先用原工程副本；短片段测试应使用没有待保留长动画的副本，不要为绕过拒绝而删除原件。

Precise character import may complete or rebuild the generated control graph. Initial migration of an old rig requires an isolated character and replacement of all existing animation from target frame 0. Partial intervals or model-scoped work require an already complete independent graph. Additive stacks, native cycles and layers shared with unrelated objects are unsupported. Work in an original project copy; use a copy without long animation to preserve for a short test, and never delete original work to bypass a rejection.

首次处理一个新模型时，先测试 30 至 60 帧，观察躯干、手指、膝盖和脚踝，再决定是否导入完整动作。窗口提示成功表示本次流程通过检查，仍需要你播放检查动作效果。

When using a new model, first test 30–60 frames and inspect the torso, fingers, knees and ankles before importing the full motion. A success message means the operation passed its checks; you should still play the animation and inspect the result.

05 选择目标模型

Select target models

Import VMD 和 Export VMD 都会先打开模型选择窗口。列表显示名称、骨骼数量及绑定信息；鼠标停在条目上可查看完整说明。binding verified / 绑定已验证表示检测到绑定证据，不表示该模型与任意动作都兼容。

Import VMD and Export VMD first open a model picker. Each row displays the name, joint count and binding information; hover over it to read the full details. “Binding verified” indicates detected binding evidence, not compatibility with every motion.

单击条目可勾选，再次单击取消勾选；可以选择多个模型。确认目标后点击 Select / 选择。对名称相同的模型，应结合 Rig 和身份说明区分，不能只凭显示名称判断。

Click a row to select it and click again to deselect it. You can choose multiple models. Click Select when the targets are correct. For models with identical names, distinguish them using the rig and identity details rather than the display name alone.

搜索框同时搜索名称与说明。确认时只提交当前筛选结果中仍可见的已选条目。需要跨多个搜索词选择时，最后清空搜索并重新核对全部勾选项，避免遗漏。

Search matches both names and details. Confirmation includes only selected rows still visible in the current filter. If you select across several searches, clear the search and check all selections before confirming to avoid omissions.

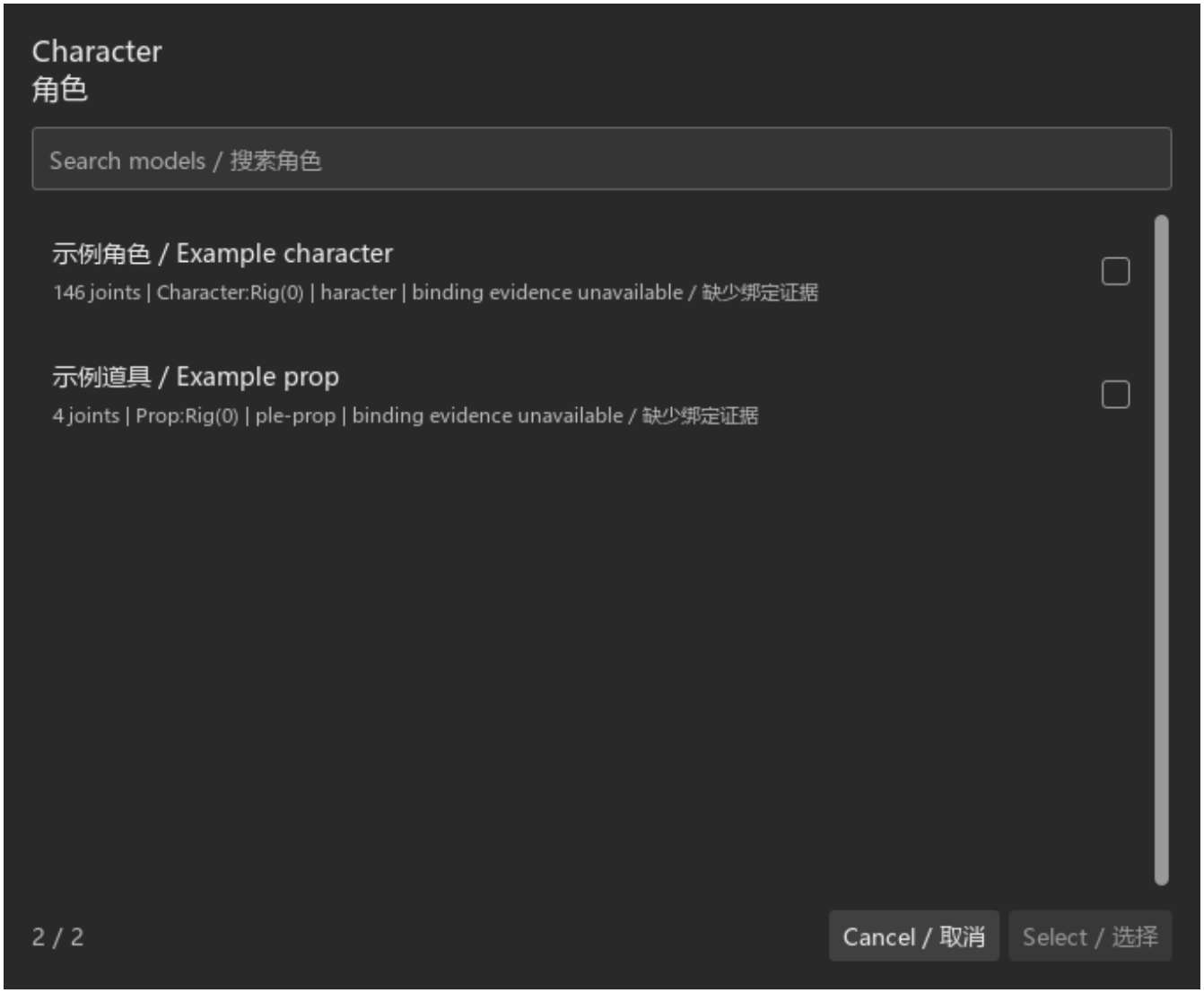
没有目标时点击 Cancel / 取消，回到场景检查骨架与绑定。不要在选择窗口或导出过程中切换工程、删除模型、重绑控制器或编辑当前帧；检测到结构变化后应关闭流程并重新开始。

If the intended model is missing, click Cancel and inspect its skeleton and bindings. Do not switch projects, delete models, rebind controls or edit the current frame while selecting or exporting. If a structural change is detected, close the operation and start again.

界面示例 / Interface example

模型信息为示意数据，请在实际工程中核对目标范围。

Model information is illustrative. Check the target scope in your own project.



目标模型与绑定信息 / Target models and binding information

06 设置时间轴范围

Set the timeline range

导入会在点击 Import VMD 时读取时间轴选区。只有移动播放头，不等于设置导入起点。没有有效选区时，源 VMD 从第 0 帧写入目标第 0 帧，一直到该动作的骨骼末帧。

Import captures the timeline selection when you invoke Import VMD. Moving the playhead alone does not set the import start. With no valid selection, VMD frame 0 is written to target frame 0 and import continues to the motion’ s final skeletal frame.

有选区时，源第 0 帧对应选区起点。例如目标选中 100–160，共 61 帧，源 0–60 被写到目标 100–160。它不是从源动作第 100 帧开始，也不会把整个动作压缩到 61 帧。

With a selection, source frame 0 maps to the beginning of the selected range. For example, target frames 100–160 contain 61 frames, so source frames 0–60 are written there. This does not start at source frame 100 or compress the full motion into 61 frames.

选区比动作短时，只导入动作开头能容纳的部分；选区比动作长时，写入范围截到动作末帧，不循环填满。请检查目标区间是否已有需要保留的动画，先用工程副本验证。

If the selection is shorter than the motion, only the initial portion that fits is imported. If it is longer, writing stops at the motion’ s end without looping to fill the selection. Check for existing animation you want to preserve and test in a project copy.

导出使用选中的帧范围；没有有效选区时，使用宿主时间轴从 0 到最后一帧的整个范围。输出 VMD 从第 0 帧重新编号。插件按 30 fps 的产品时间轴约定处理导出，没有自动侦测任意播放帧率再变速的步骤。

Export uses the selected frame range. Without a valid selection, it uses the whole host timeline from frame 0 to its last frame. The exported VMD is renumbered from frame 0. Export follows a 30 fps timeline convention; it does not automatically detect arbitrary playback rates and retime them.

如果你要从源动作中间开始，请先使用自己的动作编辑工具制作所需片段，再导入该片段。不要用目标时间轴的起始数字来代替源片段裁剪。

To start from the middle of a source motion, create the desired clip with your motion-editing tool first, then import that clip. The target timeline’ s starting number does not crop the source motion.

07 导入 VMD 动作

Import a VMD motion

1. 打开准备好的 .casc 工作副本并设置时间轴选区。依次点击 文件 → 导入 → Import VMD；英文界面对应 File → Import → Import VMD。

Open the prepared .casc working copy and set the timeline selection. Choose File → Import → Import VMD, shown as 文件 → 导入 → Import VMD in the Chinese interface.

2. 在模型窗口勾选目标，点击 Select / 选择。随后选择一个 .vmd 文件并确认。一次多选的所有模型共用这一份 VMD；如需给不同角色使用不同动作，请分次导入。

Select the target models and click Select. Choose one .vmd file in the file picker and confirm. All models selected in that operation share the same VMD. Import separately when different characters need different motions.

3. 等待处理结束，期间不要重复点击导入或操作骨架。插件会准备动作、检查绑定与目标区间，再写入控制器并读取结果核对。较长动作或多角色需要更久。

Wait for processing to finish without repeatedly invoking Import or manipulating the skeleton. The toolkit prepares the motion, checks bindings and the target interval, writes the controls, then reads the result back for verification. Longer motions and multiple characters take more time.

4. 阅读结果中的成功、失败和未执行模型列表。多模型处理中，一个失败不表示其他已成功的模型被撤回。若提示场景可能保留部分动画，后续模型会停止；保留报告，检查工作副本，不要继续叠加导入。

Read the lists of successful, failed and unprocessed models. In a batch, one failure does not mean earlier successful models were reverted. If the message warns of partial animation, remaining models are stopped. Keep the report and inspect the working copy before importing anything further.

5. 关闭结果窗口，播放检查关键部位和选区起止帧。确认满意后由你保存工程；导入流程不会代替你决定是否保存。

Close the result window, play the animation and inspect key body parts and the range endpoints. Save the project yourself when satisfied; the import operation does not decide whether to save your project.

此工具的主要范围是骨骼动作。不要把没有表情、镜头或灯光变化误判成骨骼导入失败，也不要仅凭动作文件能被读取就认定它适用于当前模型。

The primary scope is skeletal motion. Missing facial, camera or light changes do not by themselves indicate a skeletal import failure. A readable motion file is not necessarily suitable for the current model.

逐帧烘焙保留现有时间轴关键帧和区间的 IK/FK、固定状态。1.11 修复旧宿主可选接口缺失导致的导入失败，以及导入后验证额外触发求解、改变姿态并报告失败的问题。升级不会自动修改已经保存的旧动画；需要时请在原工程副本重新导入。

Baking preserves existing timeline key and interval IK/FK and fixation settings. Version 1.11 fixes missing optional APIs on legacy hosts and an extra solve during post-import verification that could change poses and report failure. Updating does not rewrite saved animation; re-import into an original project copy when needed.

08 使用 MMD 道具

Work with MMD props

道具也通过 Import VMD 和 Export VMD 使用。适用对象需要具有可识别的 MMD 根骨语义，以及模型、骨架和网格结构。任意无 Rig 网格、单个 Point 或普通 Rig 节点不会因此自动变成可用的 MMD 道具。

Props use the same Import VMD and Export VMD commands. A supported prop needs recognizable MMD root-bone semantics together with model, skeleton and mesh structure. An arbitrary unrigged mesh, a single Point or an ordinary Rig node does not automatically become a supported MMD prop.

人物与道具依据实际控制绑定关系区分，不按骨骼数量区分。带人体 AutoPosing 控制图的对象按人物处理；没有该控制图也仍需满足道具结构条件。请选择列表中的正确对象并先测试短片段。

Characters and props are distinguished by actual control bindings, not joint count. Objects with a humanoid AutoPosing control graph follow the character route. Objects without one must still meet the prop structure requirements. Select the correct item and test a short clip first.

道具 VMD 的骨名和运动含义需要与该道具一致。成功识别模型只是第一步，仍需检查位移、朝向、父级关系和导入后的实际运动。把人物动作直接用于独立道具，不会自动生成正确的持握或约束关系。

A prop VMD must use bone names and motion semantics suitable for that prop. Successful detection is only the first step: inspect translation, orientation, parenting and the resulting movement. Applying a character motion to a separate prop does not automatically create a correct grip or constraint relationship.

仅选择道具导出时会直接使用其骨骼轨道，不显示人物 FK / IK 策略选择。人物与道具一起导出仍生成各自的文件；之后应在目标软件中检查它们的相对运动。

Exporting only props uses their skeletal tracks directly and does not show the character FK/IK strategy picker. Exporting characters and props together still creates separate files. Check their relative movement in the destination application afterward.

请保留未绑定和已绑定的工程副本，尤其是在道具与人物合并或修改父级之前。识别成功不等于所有道具组合与约束配置均已验证。

Keep copies before and after binding, especially before merging a prop with a character or changing parenting. Successful detection does not establish that every prop combination or constraint setup has been verified.

09 导出骨骼动画

Export skeletal animation

1. 完成动画编辑后，停止播放并设置导出选区。点击 文件 → 导出 → Export VMD，即 File → Export → Export VMD。选择一个或多个目标模型。

After editing the animation, stop playback and set the export range. Choose File → Export → Export VMD and select one or more target models.

2. 如果选择中含有人物，按下一章选择 Weight bones / 权重骨（FK）或 IK bones / IK 骨。点击 Continue / 继续后，选择保存位置和 .vmd 文件名。

If the selection includes a character, choose Weight bones (FK) or IK bones as explained in the next chapter. Click Continue, then choose the destination and .vmd filename.

3. 等待 Sampling / 采样和 Verifying / 复核完成。导出读取场景，不会替你保存工程、改写动画或断开控制器。处理期间不要切换场景、移动当前帧、修改骨架或动画，否则应重新开始。

Wait for Sampling and Verifying to finish. Export reads the scene; it does not save the project, rewrite animation or disconnect controls. Do not switch scenes, move the current frame or edit the skeleton or animation during the operation. Restart if such changes occur.

4. 检查完成提示中的输出文件。单模型使用你选择的文件名；多模型以该名称为基础追加模型名。同批重名或已有同名文件会使用编号避让，例如 motion_Miku_2.vmd。取消或失败时，先前已经完成的模型文件可能仍保留，按结果列表确认。

Check the output files in the completion message. A single model uses the selected filename. A batch appends each model name to that base name. Duplicate or existing generated names receive a number, such as motion_Miku_2.vmd. Files completed earlier in a batch may remain after cancellation or failure; consult the results list.

5. 在目标 MMD 软件中加载对应模型和导出的 VMD，检查起止帧、脚部接触和手部运动。导出只包含骨骼动画和必要 IK 状态，不包含模型、表情、相机或灯光。

Load the matching model and exported VMD in the destination MMD application. Inspect the first and last frames, foot contact and hand motion. The export includes skeletal animation and necessary IK states, not the model, morphs, cameras or lights.

10 选择 FK 或 IK 导出

Choose FK or IK export

FK 是根据骨骼层级逐级计算姿态；IK 则通过目标骨和链约束求解腿或脚的姿态。两者是输出轨道策略，不是改变当前 Cascadeur 动画的开关。

FK computes poses along the bone hierarchy. IK solves a limb using target bones and chain constraints. These choices control the exported tracks; they do not switch the current Cascadeur animation to a different mode.

Weight bones / 权重骨（FK）导出场景中可用的完整骨骼集合，并按该策略处理 IK 状态。它适合优先保留实际骨骼运动的工作流；名称中的“权重骨”不表示只导出少量主骨。

Weight bones (FK) exports the available scene-native joint set and handles IK states for that strategy. It suits workflows that prioritize the actual skeletal motion. The label does not mean that only a few major bones are exported.

IK bones / IK 骨保留场景明确存在的 IK 轨道和状态，同时仍输出需要的其他骨骼及腿部 FK 初始信息。它不是只写一个 IK 开关，也不是只输出两个脚部目标。

IK bones preserves explicit scene IK tracks and states while retaining other required bones and the leg FK seed information. It does not merely write an IK switch or export only two foot targets.

只有满足条件时才能选择 IK。选择后还会检查链结构、初始姿态及输出重放；缺少有效链或目标时可能拒绝导出。不要通过改骨名或关闭 IK 求值来掩盖错误，应使用符合目标模型的骨架与绑定。

IK is available only when the required information is present. After selection, the toolkit also checks the chain, rest pose and output replay; missing or invalid chains or targets can still cause rejection. Correct the skeleton and bindings for the intended model rather than masking the issue by renaming bones or disabling IK evaluation.

导出采用 1 个 MMD 单位对应 8 厘米的比例。VMD 不支持缩放动画；检测到动态缩放时会停止。模型尺寸不同或目标端物理设置不同仍可能影响视觉结果。

Export uses 8 centimetres per MMD unit. VMD cannot represent animated scale, so detected scale animation stops export. Different model proportions or destination physics settings can still affect the visual result.

界面示例 / Interface example

模型信息为示意数据，请在实际工程中核对目标范围。

Model information is illustrative. Check the target scope in your own project.



FK / IK 导出选项 / FK and IK export choices

11 清理场景中的生成 Rig

Clean generated rigs from a scene

Scene Clean 用于复位并移除检测到的 MMD 模型生成控制 Rig，以便继续整理模型。它会改变工程内容；请先另存工作副本。它不是普通的缓存清理，也不是完成每次导入或导出都要做的步骤。

Scene Clean resets and removes generated control rigs from detected MMD models so you can continue preparing the model. It changes the project, so save a working copy first. It is not a cache cleaner or a required step after every import or export.

该操作面向当前场景中检测到的模型，不只针对大纲中高亮的一根骨骼。先检查当前帧与需要保留的动画，再打开 Scripts / Commands → cascadeur MMD toolkit → Scene Clean；英文界面为 Scripts。

The operation targets detected models in the current scene, not just the bone highlighted in the outliner. Check the current frame and animation you want to preserve, then open Scripts / Commands → cascadeur MMD toolkit → Scene Clean.

确认窗口列出目标模型、骨骼及 Rig 数量。点击 Yes / 执行后，先在当前帧将可用控制器复位到存储的初始姿势，再移除生成 Rig；网格、骨架和 Armature 保留。这不会把整个时间轴所有帧都转换成新的 T-pose 动作。

The confirmation lists target models and joint and rig counts. Clicking Yes first resets available controls at the current frame to their stored initial pose, then removes generated rigs. Meshes, skeletons and armatures remain. It does not turn every frame of the timeline into a new T-pose animation.

完成后阅读复位和删除数量，并检查大纲。若某个应复位的模型不能完整复位，会保留 Rig 并报告失败，但此前的姿势操作可能已经发生。不要忽略失败提示后继续工作；检查工程副本，必要时重新打开保留的原工程。

Read the reset and removal counts and inspect the outliner afterward. If a required reset cannot complete, the rigs are kept and failure is reported, but earlier pose changes may already have occurred. Do not ignore the message and continue blindly; inspect the working copy and reopen the preserved original if needed.

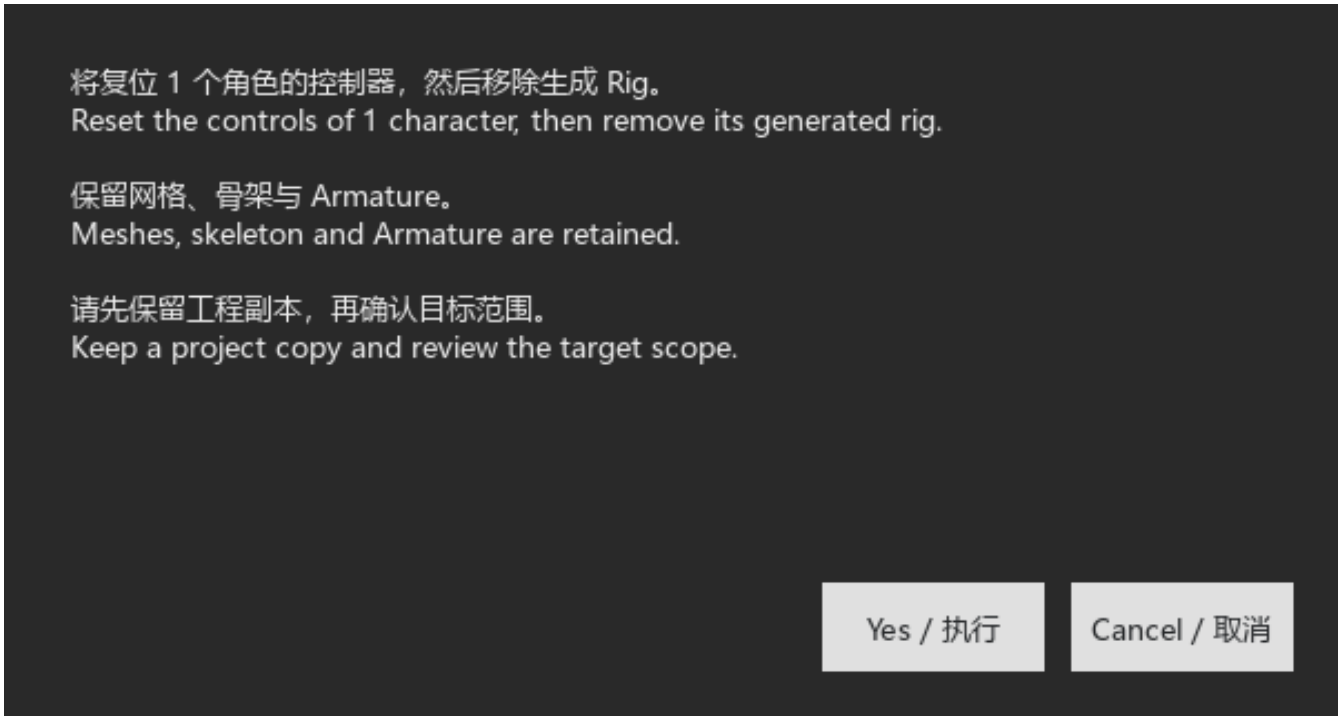
窗口会提示可使用 Undo；重要工程仍应保留独立副本。清理后的模型失去生成控制 Rig，若要再次通过人物控制器导入动作，需要重新完成相应绑定流程。

The window mentions Undo, but keep a separate copy of important projects. Cleaning removes generated controls. Importing through character controls again requires rebuilding the appropriate rig and bindings.

界面示例 / Interface example

模型信息为示意数据，请在实际工程中核对目标范围。

Model information is illustrative. Check the target scope in your own project.



清理前确认目标范围 / Review the target scope before Scene Clean

12 查找报错原因

Diagnose an error

菜单没有出现：确认安装成功、启动了正确目录下的 Cascadeur，并完全退出旧进程后重新打开。打开或切换到角色工程后再检查菜单。若仍缺失，保留安装提示与 toolkit-native.log，不要反复手动复制部分文件。

Menus are missing: confirm installation succeeded, launch Cascadeur from the correct directory and fully restart the old process. Open or activate a character project and check again. If the menus remain missing, keep the installer message and toolkit-native.log instead of repeatedly copying individual files.

No MMD model was detected / 未检测到 MMD 模型：确认打开的是预期工程，人物具有有效绑定，道具具有 MMD 骨架与模型结构。只修改显示名称或增加一个 Rig 节点不能补齐结构。

No MMD model was detected: confirm that the intended project is open, characters have valid bindings and props have the required MMD skeleton and model structure. Changing a display name or adding a Rig node cannot supply missing structure.

导入失败或运动不正确：先查看报告中的失败模型、帧号与错误类型。读回误差、绑定变化、非法数值和源动作不兼容需要分别排查。VMD 不携带完整模型骨架定义，因此文件未损坏不等于动作语义适用于当前模型。

Import fails or motion looks wrong: read the report’s model, frame and error type first. Readback errors, changed bindings, invalid numbers and incompatible source motions have different causes. VMD does not contain a complete model skeleton definition, so an intact file can still be semantically unsuitable for the target model.

导出失败：检查是否在处理期间编辑过场景，是否存在动态缩放，IK 模式下是否有有效的链和目标。长骨名受 VMD 的日文编码和 15 字节骨名字段限制，无法无歧义区分的名称可能被省略，应阅读报告警告。

Export fails: check for scene edits during processing, animated scale, and valid chains and targets in IK mode. Long bone names are constrained by VMD’s Japanese encoding and 15-byte bone-name field. Ambiguous names may be omitted, so read the report warnings.

导出提示“已按场景骨架导出”时，说明候选 PMX 的父骨等信息不能与当前工程完整对应。输出使用场景骨架，报告的 target_contracts 会记录原因。这不表示原 PMX 的全部辅助骨、物理骨或自定义约束已获验证；请在目标模型上检查重放。需要原 PMX 语义时，应先使用骨架一致的工程与 PMX。

If export reports that the scene skeleton was used, the candidate PMX parent graph or other metadata could not be fully matched to the project. The output uses the scene skeleton and target_contracts records the reason. This does not verify every helper bone, physics bone or custom constraint in the original PMX. Check playback on the target model. Use a matching project and PMX when original PMX semantics are required.

报告和日志位于 %LOCALAPPDATA%\cascadeur MMD toolkit。导入报告在 bake_work，导出报告在 export_reports；主要日志为 toolkit.log、toolkit-native.log 和 toolkit-flow-trace.jsonl。错误窗口给出的具体报告路径优先。

Reports and logs are under %LOCALAPPDATA%\cascadeur MMD toolkit. Import reports are in bake_work and export reports in export_reports. Main logs are toolkit.log, toolkit-native.log and toolkit-flow-trace.jsonl. Prefer the exact report path shown in the error window.

反馈问题时提供完整报错、对应报告、插件版本、Cascadeur 版本，以及可复现问题的工程和动作副本。工程与动作可能含个人路径或未公开素材，请先确认可分享范围。

When reporting an issue, include the complete error, its report, the toolkit and Cascadeur versions, and project and motion copies that reproduce the problem. Check what you can share, since projects and motions may contain personal paths or unpublished assets.

13 完成一次完整工作流

Complete a full workflow

1. 安装并重新启动 Cascadeur，打开角色工程副本，确认模型选择窗口能识别正确对象。

Install, restart Cascadeur, open a copy of the character project and confirm that the model picker identifies the intended object.

2. 先选择短时间轴范围，导入一份与模型适配的 VMD。关闭结果提示，检查躯干、手指、膝盖、脚踝与首尾帧。

Select a short timeline range and import a VMD suitable for the model. Close the result message and inspect the torso, fingers, knees, ankles and range endpoints.

3. 在工作副本中完成动作编辑，选择输出范围，按目标需求选择 FK 或 IK，导出到独立的动作文件夹。

Finish editing in the working copy, select the output range, choose FK or IK for the destination workflow and export to a dedicated motion folder.

4. 在目标软件用对应模型重放输出，确认速度、尺寸、脚部接触与手部运动。需要保留的工程由你保存；无需清理时跳过 Scene Clean。

Replay the output on the matching model in the destination application. Check timing, scale, foot contact and hand motion. Save the project you want to keep and skip Scene Clean unless you need its rig-removal function.

5. 保留原工程、完成的工程、源 VMD、输出 VMD 和安装备份。这样即使以后调整绑定或升级插件，也能回到明确的工作阶段。

Keep the original project, finished project, source VMD, exported VMD and installation backup. These let you return to a known stage after changing bindings or updating the toolkit.

本教程对应 cascadeur MMD toolkit 1.11。旧导入插件 v1.1 / v1.3 是各自独立的历史版本，菜单、安装方式和功能范围应以各自说明为准。

This guide describes cascadeur MMD toolkit 1.11. The earlier importer v1.1/v1.3 releases are separate historical versions; use their own documentation for their menus, installation and feature scope.